

Sebastian Mostek

Technical Portfolio

October 2026

1 Introduction

This document serves to record and present my technical skills. A more interactive and detailed version of mostly the same information can be found on [my website](#).

2 MATLAB

MATLAB is the programming language I am most familiar with and has been the workhorse behind the successful implementation of various research, academic, and personal projects.

2.1 Independent Mathematics Research

Beginning as a mere curiosity, the question of how electrons would distribute to reach equilibrium on a sphere would result in several semesters of independent study and a great deal of experience in advanced math and programming. The full explanation of the problem and the code I used to investigate it can be accessed through [my GitHub page](#).

One of the more programmatically elaborate scripts to come out of this project was an interactive app used for visualizing the potential energy distribution. Essentially, one can imagine each of the points in the diagram as electrons, and the contour lines as representing potential energy well of the blue point. The user can click and drag that point to move it around the sphere, but the confusing part is that the other points — those creating the potential well — also move. This is because the program preserves the symmetry of the original shape: each of the yellow points in Figure (1) is reached by reflecting the blue point across the dashed lines.

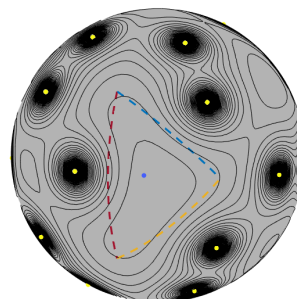


Figure 1: UI for potential energy visualization app

2.2 Dynamics Simulations

The most technically impressive simulation I've ever programmed was of a ball rolling within a hemispherical shell that itself can roll. There is no one aspect of the development process that I can point to as the primary technical or conceptual hurdle of that project, rather it was the integration of dozens of small challenges that combined into such a challenging whole.

More striking perhaps is the simulation, designed for *MAE 3260 System Dynamics*, of a satellite using reaction wheels to control its orientation. The focus was on how the linear control laws interacted with the nonlinear equations of motion.

These are not systems that a single image can do justice, but [my website](#) displays the animations as gifs, and I also have a [github page](#) to store the code.

3 AutoDesk Fusion

The design files I inherited upon starting at Private Machines were, put simply, a complete mess. I learned much about good design practice and file management from making note of what didn't work well with what we already had.

In my short tenure I would redesign all of the machined and sheet metal components from scratch. The resulting single design file for our server chassis is shown in figure 3 offered much improved editability and

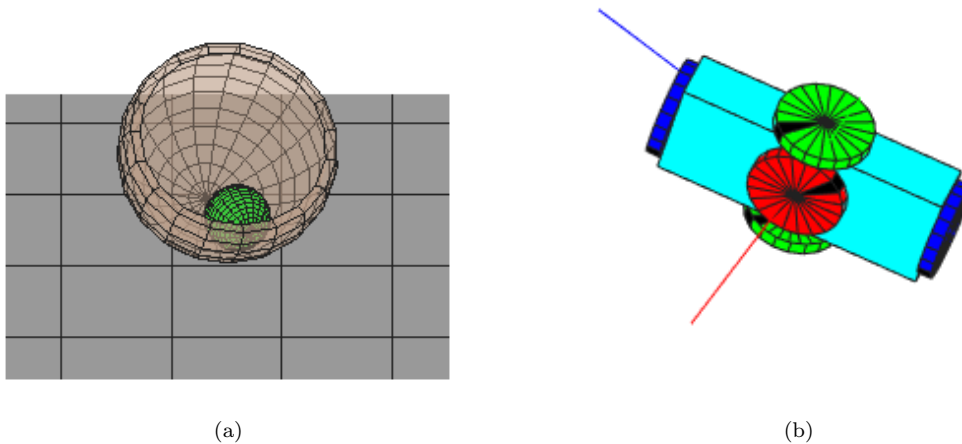


Figure 2: (a) Simulation of a ball rolling within a larger rolling shell. (b) Simulation of a satellite adjusting to a new orientation

also featured a significant improvement in terms of DFM and ease of assembly – I was able to remove nearly half of the total fasteners.

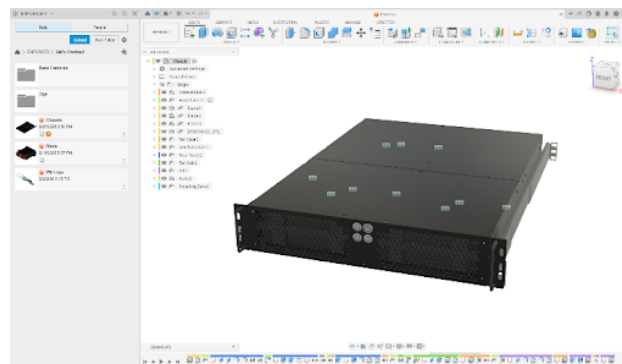


Figure 3: The entire server chassis designed in a single file, offering a significant increase in editability.

4 Ansys

4.1 Discovery

Though we used Fusion for CAD at Private Machines, there was a need for CFD simulation and it was cheaper to get an Ansys license than an Autodesk CFD license.

Discovery's ability to quickly iterate through variations on combined heat/air-flow simulations would prove crucial to the experiments I ran to help optimize the cooling of our servers. The high-security design of our servers required about 4 inches of aluminum and 2 thermal interfaces between the cpu and the outside air. Those would prove to be a significant problem as my simulations and mathematical models suggested we were gaining nearly 30°C in cpu temperature due to poor thermal contact.

4.2 Fluent

The final project of *MAE 4021 Wind Power* involved the performance evaluation of a given wind turbine blade and the improvement on that design. This required running hundreds of CFD simulations in Ansys

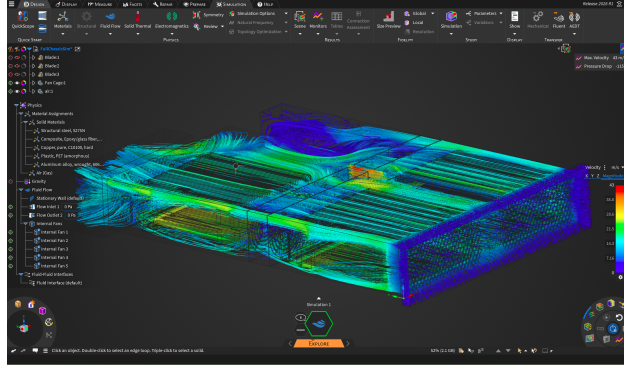


Figure 4: Simulation of airflow through the server chassis.

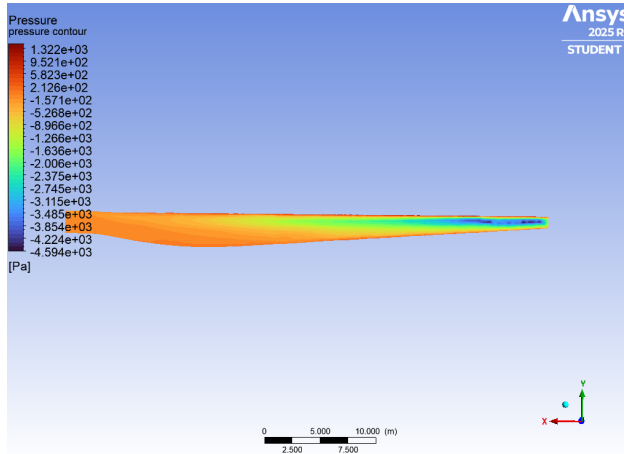


Figure 5: pressure contour on downstream face of a model wind turbine blade.

Fluent at different wind speeds, rotation rates, and blade pitch angles.

While this Ansys experience was valuable, the more notable aspect of this project was the unique derivation I used in the design portion. Essentially, I was able to identify and rectify a flaw in the mathematical argument made by the course textbook, and the result was a far simpler method of predicting blade performance that provided greater agreement with reference data than did the CFD simulation. The full details can be found in my report submission via [my website](#).